

attribute (where one can understand what the attribute is for).

- SYNTAX 1.3.6.1.4.1.1466.115.12.1.1.25 – The 'SYNTAX' keyword defines the syntax, or in other words, the format of this attribute—whether numerical, string, etc. The syntax is defined in terms of Syntax OIDs again. The list of these OIDs can be referenced from the rfc (<http://www.rfc-editor.org/rfc/rfc2252.txt>).

Let us look at another example – that of the attribute type 'sn' or SurName. This attribute would contain the surname of any person, for example, 'Tendulkar' for 'Sachin Tendulkar' or 'Gupta' for 'Varad Gupta'. The requirements for this would be character type data (*char* or *varchar* for database admins/developers), since normally no human would have a surname with numerals.

```
[root@bvg ~]# cat /etc/ldap/schema/core.schema | \
grep -A 5 'attributetype ( 2.5.4.4 NAME ( 'sn' 'surname' ) )'

attributetype ( 2.5.4.4 NAME ( 'sn' 'surname' )
    DESC 'RFC2256: last (family) name(s) for
which the entity is known by'
    SUP name )
```

It's important to note the following:

- `attributetype ( 2.5.4.4 NAME ( 'sn' 'surname' )` – This attribute has two names or, in other words, it has an alias. It can be referred to as 'sn' or 'surname' in a record.
- `SUP name` – This specifies that this attribute has a superior attribute (specified by the keyword - SUP) called 'name'. Like class hierarchies (OOPs - for programmers), attributes also have hierarchies.

This definition specifies that the attribute 'sn' will inherit the properties of its superior attribute, or in other words 'sn' is derived from 'name'.

The superior attribute for 'sn' is 'name'. `name` is a system (inbuilt) attribute in OpenLDAP. Its definition, though, is depicted in the *core.schema* file, but commented. The attribute name is defined thus:

```
#attributetype ( 2.5.4.13 NAME 'description'
# DESC 'RFC2256: descriptive information'
# EQUALITY caseIgnoreMatch
```

```
# SUBSTR caseIgnoreSubstringsMatch
# SYNTAX 1.3.6.1.4.1.1466.115.12.1.15{
1024} )
#
```

The syntax for the name attribute is defined as:

```
SYNTAX 1.3.6.1.4.1.1466.115.12.1.15{1024}
```

The Syntax OID 1.3.6.1.4.1.1466.115.12.1.15 means a directory string and {1024} specifies the maximum length. Therefore, the 'name' attribute can be a string of up to 1,024 characters. Since the 'sn' attribute is derived from its SUPERIOR attribute 'name', it too can be a string of up to 1,024 characters.

We will cover the EQUALITY and SUBSTR clauses later when we deal with indexes. I hope that you can now figure out how data types are defined and used in OpenLDAP. Having seen how attributes (or columns in an individual sheet) are defined, in this article, we will probe deeper into ObjectClasses (individual sheets of a spreadsheet file, going by our analogy) in the next article.

It is left as an exercise to the reader to understand the attributes used in our sample of an Ldif file given above—'dc', 'ou', 'cn', 'l', 'street', 'st', 'postalCode', 'homePhone', 'mobile' and 'mail'. Good documentation to understand attributes and ObjectClasses is available at <http://technology.artifact-software.com/LDAPSchema/ldapschema.html>.

Do feel free to drop me a mail in case you have any doubts or run into difficulties while understanding the hierarchy and/or syntax of these attributes. 

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